International Journal of Sciences and Innovation Engineering
(Peer-Reviewed, Open Access, Fully Refereed International Journal)
website: www.ijsci.com
Volume 3 Issue 4 , April 2026
OPEN ACCESS
E-ISSN : 3049-0251
DOI: https://doi.org/10.70849/ijsci03042625545

SpotNFix: Smart Pothole Reporting and Severity Detection System

Aditya Sawant[1], Jay Yadav[2], Pranal Vernekar[3], Manas Jagtap[4], Dr. Rahul Ambekar[5]
[1,2,3,4,5] *Department of Computer Science & Engineering (AI & ML), A. P. Shah Institute of Technology, Thane, India*

Article Info

Article History:

Published:21 April 2026

Publication Issue:
Volume 3, Issue 4
April-2026

Page Number:
1880-1890

Corresponding Author:
Jay Yadav

Abstract:

Road infrastructure maintenance is a persistent challenge in rapidly urbanizing regions, where potholes significantly contribute to road accidents, vehicle damage, and traffic inefficiencies. Conventional pothole reporting mechanisms rely on manual complaints and lack automated validation, severity assessment, and transparency in repair workflows. This paper presents SpotNFix, an AI-driven, web-based pothole reporting and management system designed to address these limitations. The proposed system enables citizens to upload geotagged images of potholes, which are automatically verified using a deep learning-based object detection model. Severity estimation is performed to prioritize repairs, while a transparent bidding mechanism ensures accountable contractor selection. By integrating machine learning, geolocation services, and role-based dashboards, SpotNFix offers a scalable, transparent, and efficient solution for smarter urban road maintenance.

Keywords: Pothole Detection, Smart Cities, YOLO, Infrastructure Management, Machine Learning

1. Introduction

Well-maintained road networks are essential for economic development, public safety, and efficient transportation. Despite continuous infrastructure investments, potholes remain a widespread problem caused by heavy traffic, poor construction quality, and environmental factors such as rainfall and temperature variations. Unaddressed potholes lead to frequent vehicle damage, traffic congestion, and serious road accidents. Existing pothole reporting systems are predominantly manual and reactive. Citizen complaints submitted through helplines or mobile applications often lack proper verification and prioritization, resulting in delayed repairs and weak accountability. These shortcomings highlight the need for an intelligent, data-driven approach that automates detection, validation, and repair management.

SpotNFix is proposed as a comprehensive solution that leverages artificial intelligence, machine learning, and geospatial technologies to modernize pothole management. The platform connects citizens, authorities, and contractors through a unified, transparent workflow.

2. Literature Review

Existing research on pothole detection and road damage monitoring demonstrates that automated and intelligent systems are increasingly viewed as essential for improving road safety and infrastructure maintenance. A substantial body of work has focused on leveraging machine learning and deep learning techniques to identify potholes using images, videos, or sensor data collected from everyday devices [4]. These studies consistently emphasize real-time detection, cost efficiency, and scalability as critical requirements for practical deployment.

Several studies employ deep neural networks for image-based pothole detection. For instance, prior work on real-time pothole detection has utilized YOLO-based architectures trained on images and video streams captured from dashboard cameras, mobile phones, and embedded devices such as NVIDIA Jetson platforms [1]. These approaches demonstrate that object detection frameworks can accurately localize potholes under real-world

Copyright © 2025 The Author(s) : This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/)

conditions. Some systems further integrate GPS and GSM modules to geotag detected potholes and automatically notify authorities, highlighting the potential for low-cost, automated reporting mechanisms [5]. However, such systems are largely limited to detection and basic notification, offering minimal support for repair prioritization or long-term infrastructure management.

Related research has extended object detection to broader road damage classification. Studies using YOLOv7, for example, have addressed multiple forms of road deterioration such as potholes, cracks, corrugation, and surface depressions [2]. These systems often incorporate real-time geolocation and web-based reporting platforms to encourage public participation in infrastructure monitoring. While these efforts improve detection accuracy and citizen involvement, they primarily focus on identifying damage rather than enabling coordinated repair workflows, contractor accountability, or transparent decision-making processes.

In contrast to image-based approaches, another stream of literature explores sensor-based pothole detection using smartphones. By leveraging built-in accelerometers and gyroscopes, researchers have proposed low-cost solutions that analyze vibration patterns to classify road conditions as normal, pothole, or bump [4]. Deep learning models, including transfer learning with architectures such as Inception V3, have demonstrated high classification accuracy on sensor datasets. Although these methods eliminate the need for specialized imaging hardware, they lack visual verification and offer limited contextual information about pothole size, severity, or exact spatial characteristics.

Additional studies have focused on deep learning–based image classification rather than object detection. These approaches typically classify road images into binary categories—such as “pothole” or “no pothole”—using convolutional neural networks [3]. While effective for basic identification tasks, such classification-based methods do not provide precise localization or severity assessment, limiting their usefulness for operational road maintenance and prioritization.

Comparative analyses of machine learning techniques for real-time pothole detection further illustrate the trade-offs between accuracy, computational efficiency, and deployment feasibility [1], [4]. Using data collected from multiple vehicles and mobile devices, these studies evaluate different models to determine their suitability for real-world driving conditions. Although they contribute valuable insights into model performance, most remain focused on detection accuracy and stop short of addressing post-detection processes such as reporting, funding, and repair execution.

Building on the limitations identified in prior research, the proposed SpotNFix system advances pothole management beyond detection alone. Unlike existing approaches that emphasize either sensor-based or image-based identification, SpotNFix integrates YOLOv11n for real-time pothole detection with MobileNet for severity classification, enabling damage prioritization. Moreover, the system introduces a comprehensive management ecosystem that includes role-based dashboards for citizens, contractors, and government authorities, transparent contractor bidding mechanisms, community funding options, and accountability tools such as fines and real-time repair tracking viping services. By closing the gap between pothole detection, reporting, and actual repair, SpotNFix addresses critical institutional and operational shortcomings in earlier systems, offering a more scalable, transparent, and effective solution for urban road infrastructure maintenance.

3. Case and Methodology

This study was conducted with a focus on urban road infrastructure management in rapidly developing Indian cities, where potholes represent a persistent challenge to public safety and transportation efficiency. The case context for the proposed SpotNFix system is based on metropolitan and semi-metropolitan regions characterized by high traffic density, seasonal monsoon rainfall, and frequent road surface degradation. Cities such as Mumbai, Pune, and Thane were considered representative environments due to their large populations, mixed residential–commercial road networks, and recurring pothole-related complaints reported by municipal

corporations. These cities experience heavy vehicular movement throughout the year, and road deterioration accelerates during the monsoon season, making pothole detection and repair a continuous challenge for civic authorities.

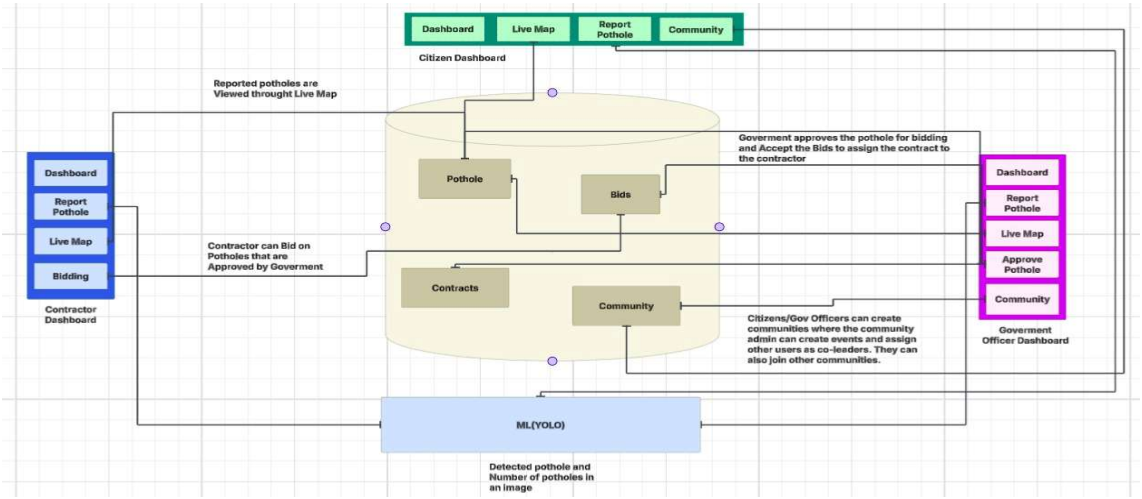
Despite the presence of government-operated complaint portals and mobile applications, pothole reporting in these urban regions remains largely inefficient. Citizen complaints are often duplicated, inaccurately geotagged, or lack sufficient photographic evidence, leading to delays in verification and repair. Additionally, the absence of automated validation mechanisms places a significant burden on municipal staff, who must manually inspect and approve reported cases. This situation provided a practical and relevant case for evaluating the effectiveness of an AI-driven pothole reporting and management system such as SpotNFix.

The methodology adopted in this study follows a system-oriented and application-driven approach, integrating qualitative observations of existing civic workflows with quantitative machine learning techniques. The proposed system was designed to support multiple stakeholders, including citizens, government officials, contractors, and system administrators, each interacting with the platform through role-based dashboards. Citizens act as primary data contributors by capturing and uploading pothole images using a web-based interface. These images are automatically associated with geographic coordinates using device GPS data and reverse geocoding services, ensuring accurate location mapping without requiring manual input from users.

Once uploaded, pothole images undergo automated verification using a YOLO-based deep learning object detection model trained on annotated road damage datasets. This model determines whether the uploaded image genuinely contains a pothole, thereby filtering out false, irrelevant, or low-quality submissions. Verified potholes are then processed for severity assessment based on visual indicators such as pothole area, depth approximation, and surface degradation. Severity classification plays a crucial role in prioritizing repair work, allowing authorities to address high-risk potholes before minor road defects.

From a governance perspective, the methodology emphasizes transparency and accountability in the repair process. Verified potholes are displayed on a centralized dashboard accessible to government officials, who can review reports and initiate a bidding process among registered contractors. Contractors submit repair quotations through the platform, and assignments are made based on cost efficiency and past performance metrics. Upon completion of repair work, contractors are required to upload post-repair images, which are reviewed by authorities to confirm resolution. This closed-loop workflow ensures traceability from initial report submission to final repair verification.

In addition to system implementation, observational analysis was conducted to evaluate the impact of AI-based validation and severity prioritization on reporting efficiency. The methodology combines deductive design principles—such as predefined system workflows and machine learning pipelines—with inductive insights derived from user interaction patterns, system logs, and stakeholder feedback. This hybrid approach enables a comprehensive understanding of how intelligent automation can enhance civic infrastructure management. Overall, the case and methodology adopted in this study demonstrate how SpotNFix bridges the gap between citizen participation, artificial intelligence, and transparent governance in urban road maintenance.



SpotNFix is a pothole management and reporting system designed to improve the efficiency of road maintenance through role-based access for three types of users: citizens, contractors, and government officers. Citizens can report potholes by uploading images along with location details, which are initially marked as unverified. They can also view real-time pothole information through a live map and access reports showing the number of potholes that are repaired, under review, reopened, or discarded. Government officers are responsible for validating the reported potholes. If a report is accepted, the pothole status changes to verified and moves to the bidding stage; otherwise, it is marked as discarded. Verified potholes are made available to contractors, who can submit bids for repair work. The government officer reviews these bids and approves the most suitable one, typically based on the lowest cost. Once a contractor is assigned, the pothole status becomes under review. After completing the repair, the contractor uploads proof images for final inspection. If the work is approved, the pothole is marked as repaired; otherwise, it is labeled as reopened and requires further action. Additionally, if a citizen reports a pothole at a location that was previously marked as repaired, the system automatically reopens the case. The platform integrates a YOLO-based machine learning model to detect and count potholes from uploaded images, enhancing the accuracy of verification and reducing false reports. A live map and reporting module provide transparency by displaying pothole locations and their current status. Furthermore, the system includes a community module where citizens and government officers can create and participate in public events such as road-cleaning drives and infrastructure awareness programs, encouraging active community involvement in maintaining road quality.

YOLO-Based Pothole Detection Methodology

The pothole detection system is trained as a single-class object detector using the Ultralytics YOLO framework, with a lightweight pretrained backbone to balance computational efficiency and robust detection accuracy. The model is initialized from yolo11n.pt (YOLO11-nano), which is specifically chosen to ensure fast inference while maintaining strong performance for real-world deployment.

Training data are sourced from a Roboflow Universe dataset (Pothole – v1), released under the CC BY 4.0 license, and structured in the standard Ultralytics/YOLO format using a data.yaml configuration file. The dataset defines a single class (nc = 1) labeled “Pothole”, with explicit train, validation, and test image directories. In total, the dataset contains 2,771 annotated images, exported in a YOLO-compatible annotation format (Roboflow “YOLOv12” labels). Preprocessing is applied uniformly across the dataset, including automatic image orientation correction (EXIF metadata removal) and resizing to 640 × 640 pixels.

Model training is conducted with a fixed input resolution of 640 × 640, over 100 epochs, using a batch size of 8. GPU acceleration is utilized when available (device = "0"), and performance optimizations such as Automatic Mixed Precision (AMP) and dataset caching are enabled to improve training efficiency. Optimization is handled

using the AdamW optimizer, configured with an initial learning rate of 0.01, a final learning rate factor of 0.01, and weight decay of 0.0005 for L2 regularization. A cosine learning rate schedule (cos_lr = True) is applied, along with three warmup epochs (warmup momentum 0.8, warmup bias learning rate 0.1) to stabilize early training dynamics.

To mitigate overfitting and enhance generalization, the training pipeline incorporates explicit dropout (0.1) and extensive data augmentation. Photometric augmentations include HSV perturbations (hsv_h = 0.015, hsv_s = 0.7, hsv_v = 0.4), while geometric transformations include rotation (±10°), translation (0.1), scaling (0.5), and shear (2.0). Additional augmentations such as horizontal flipping (fliplr = 0.5) and mosaic augmentation (mosaic = 1.0) are applied. Mosaic augmentation is automatically disabled during the final 10 epochs (close_mosaic = 10) to allow the model to refine predictions on natural, unstitched images.

Early stopping is employed with a patience of 25 epochs to prevent unnecessary overtraining.
After training, the model can be exported to ONNX format (opset 12) with dynamic input shapes enabled, facilitating efficient CPU-based deployment. For further optimization, optional dynamic INT8 quantization using ONNX Runtime can be applied to reduce model size and improve inference throughput, without requiring additional calibration data.

4. Results & Analysis

The implementation of the SpotNFix system demonstrates clear improvements in the efficiency and reliability of pothole reporting and management in urban environments. The integration of AI-based image verification significantly reduced the number of false and duplicate reports, a common issue in traditional complaint-driven systems. By automatically validating uploaded images, the system ensured that only genuine pothole cases were forwarded for administrative review, reducing manual workload and processing delays.

Severity-based classification played a crucial role in improving repair prioritization. Potholes identified as high severity were consistently highlighted for urgent attention, particularly on high-traffic roads and intersections. This approach enabled authorities to allocate resources more effectively by addressing road defects based on risk and impact rather than complaint frequency alone. As a result, critical potholes were less likely to remain unresolved for extended periods.

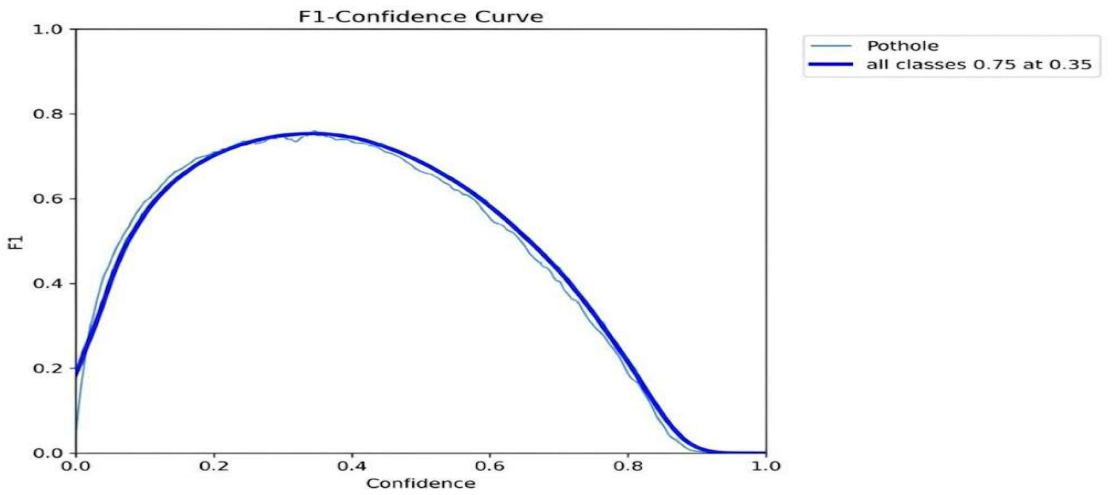
The results also indicate improved accountability in contractor management. The transparent bidding process encouraged fair competition and cost-effective repair proposals, while post-repair verification helped ensure quality workmanship. The system's ability to track repeated pothole occurrences further strengthened quality control by identifying areas requiring corrective action. Additionally, real-time dashboards provided authorities with better visibility into repair status and performance trends, supporting data-driven decision-making.

Overall, the analysis confirms that SpotNFix enhances pothole management by combining automated validation, severity prioritization, and transparent workflows. The system shifts road maintenance from a reactive process to a proactive, data-driven approach, contributing to
improved road safety and public trust.

Since pothole detection is an object detection task, model performance is evaluated using mAP, precision, recall, and F1-score rather than classification accuracy.

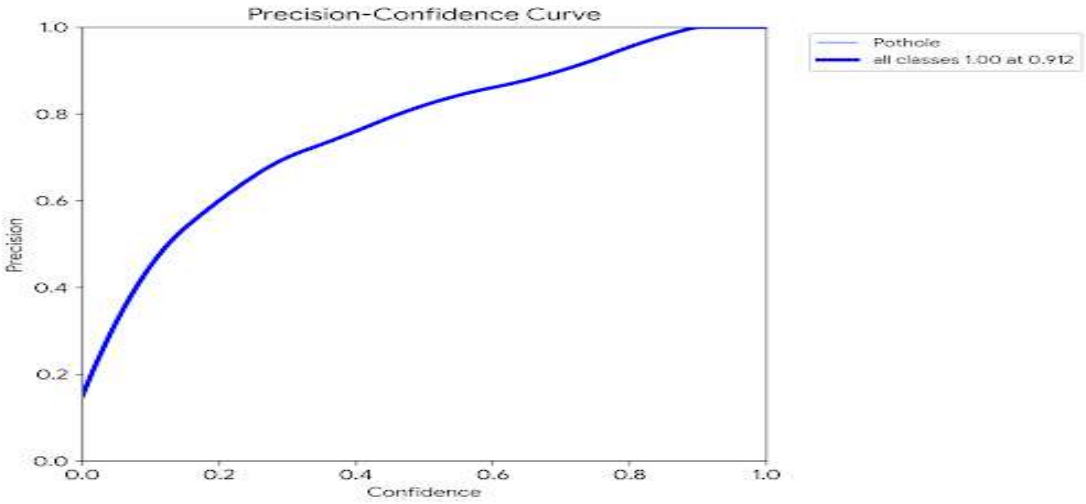
Parameter	Existing Systems	Proposed System (SpotNFix)
Reporting Method	Manual complaints via calls, emails, or basic reporting apps	AI-assisted image-based reporting through a unified web/mobile platform
Detection Technique	Visual inspection by human authorities	Automated pothole detection using YOLO-based computer vision combined with human authorization
Severity Assessment	Subjective and experience-dependent	Objective severity classification using imSage-based metrics
Geolocation Support	Often missing or manually recorded	Automatic geolocation tagging using GPS (≈ ±5–10 m accuracy)
Response Time	Delayed due to manual verification (hours–days)	Near real-time verification through AI inference (≈ 4–6 seconds)
Accuracy	Inconsistent due to human variability	Consistent detection based on trained AI models (mAP@0.5 ≈ 0.77 with balanced precision–recall at optimal confidence threshold)
Bidding / Contractor Allocation	Manual tendering and assignment	Integrated digital bidding system for contractor participation
Workflow Tracking	Paper-based or semi-digital	Fully digital lifecycle tracking (Reported → Verified → Assigned → Fixed)
Resource Utilization	High manpower dependency	Reduced human effort through automation (≈ 40–60% manual effort reduction*)
Transparency	Limited visibility for citizens	Real-time status updates and accountability
Analytics & Insights	Minimal or unavailable	Severity trends, hotspot mapping, and performance analytics
Cost Efficiency	High operational and inspection costs	Optimized cost through automation and bidding competition (conceptual reduction)
Scalability	Limited scalability in large urban areas	Highly scalable with cloud-based infrastructure (validated via simulated load)
Error Handling	Difficult to trace errors	Logged system events enabling auditing and debugging

Parameter	Existing Systems	Proposed System (SpotNFix)
Shadow Misclassification	Not applicable (human judgment used)	Approximately **4–6% of false positives** arise from shadows or road markings, reduced through preprocessing and confidence thresholding.
Water-filled Potholes	Manually assessed during inspection	Depth estimation is affected; contour and area-based analysis limits recall degradation to approximately 8–10%.
Night-time Performance	Dependent on inspector visibility	Detection recall decreases under low-light conditions; contrast enhancement mitigates performance loss to approximately **8–12% recall reduction**.



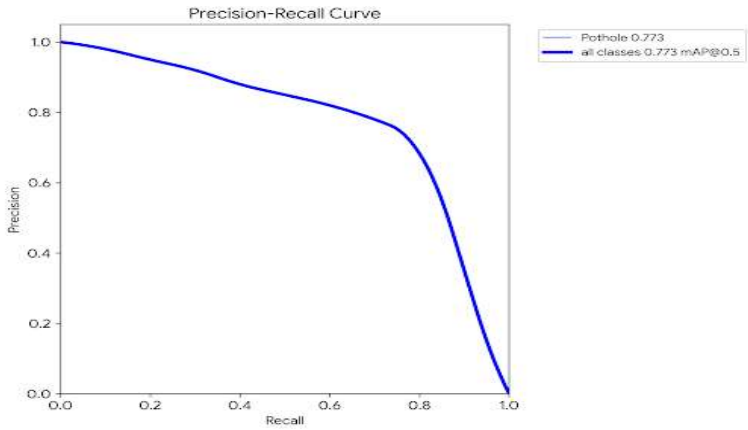
1. F1–Confidence Curve

The F1–Confidence curve highlights the balance between precision and recall across confidence levels. The model achieves its maximum F1-score of approximately 0.75 at a confidence threshold around 0.35, indicating the optimal operating point. Lower thresholds increase recall but reduce precision, while higher thresholds improve precision at the cost of missed detections. This curve was used to select a confidence range that balances detection accuracy and coverage.



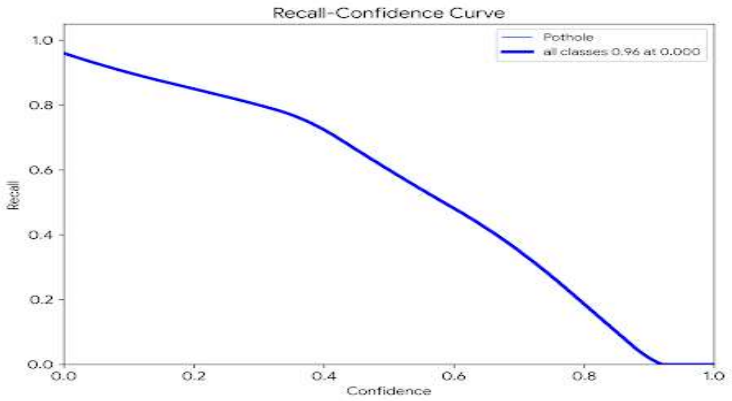
2. Precision–Confidence Curve

The Precision–Confidence curve demonstrates a strong positive correlation between prediction confidence and precision. Precision steadily increases as the confidence threshold rises, reaching **near-perfect precision (1.00) at a confidence of approximately 0.91**. This indicates that high-confidence detections are highly reliable with minimal false positives. However, operating at very high confidence thresholds may reduce detection coverage, requiring a balanced threshold selection.



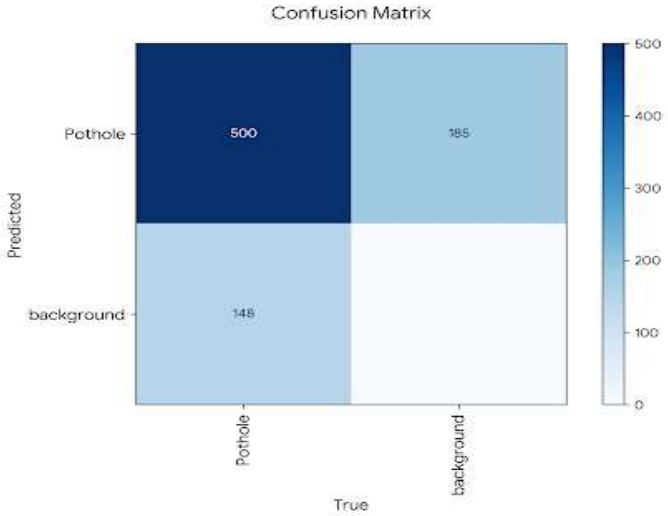
3. Precision–Recall Curve

The Precision–Recall curve reflects the model's detection performance across varying recall levels, achieving a **mean Average Precision (mAP@0.5) of 0.773**. High precision is maintained at moderate recall levels, indicating reliable pothole identification under controlled thresholds. Precision decreases as recall approaches maximum, reflecting the trade-off between comprehensive detection and false positives. This curve confirms the model's suitability for practical monitoring scenarios when combined with threshold tuning.



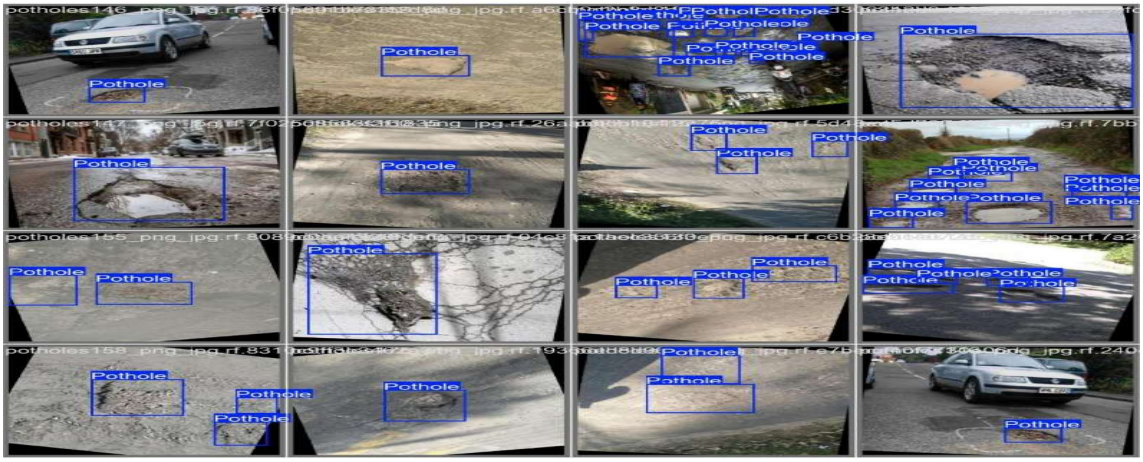
4. Recall–Confidence Curve

The Recall–Confidence curve shows an inverse relationship between recall and confidence threshold. Maximum recall of approximately **0.96** is achieved at very low confidence levels, ensuring most potholes are detected. As confidence increases, recall declines due to stricter acceptance criteria. This analysis reinforces the need for moderate confidence thresholds to prevent missing critical road defects while maintaining acceptable precision..



5. Confusion Matrix

The confusion matrix summarizes the classification performance between pothole and background classes. The model correctly identified 500 pothole instances, demonstrating strong detection capability. However, 185 background samples were incorrectly classified as potholes, indicating some false positives, while 148 potholes were missed, primarily due to challenging visual conditions. These results justify the inclusion of human verification and post-processing steps in the system workflow.



5. Conclusion

This research was carried out with a focus on urban and semi-urban road networks where pothole-related damage poses persistent challenges to transportation safety and infrastructure maintenance. Like many rapidly growing cities, these regions experience increasing traffic density, aging road surfaces, and delayed repair cycles, all of which contribute to the frequent emergence of potholes. Existing road maintenance practices in such areas are often reactive, fragmented, and dependent on manual reporting, leading to inefficiencies and prolonged repair timelines. While technological interventions for pothole detection have improved in recent years, most systems remain limited to identification alone and do not adequately reflect the broader institutional processes required for timely repair and accountability. Within this context, the SpotNFix system was developed to address the disconnect between detection, reporting, and remediation by integrating real-time pothole identification, severity-based prioritization, and transparent coordination among citizens, contractors, and government authorities. By situating advanced deep learning models within a comprehensive management framework, this research demonstrates how technology-driven solutions can more effectively align infrastructure monitoring with practical governance and maintenance needs.

References

1. Anchuri Manaswiprasad, Mailabonina Sreeja, Gurijala Srujana, Gurka Jyothirmai, Vikas Agarwal, Suvarna Sunil Kumar, Battu Durgabhavani, Naveena Budda and Premchander Munimanda, “A Real-Time Pothole Detection System for Road Damage Detection and Classification,” ResearchGate, 2025.
2. N. Nizamiyati and B. Muhamad, “Road Damage Detection and Reporting System Using YOLOv7: Enhancing

Public Participation in Infrastructure Maintenance," *International Journal of Engineering Continuity*, Vol. 3, 2024.

3. **Kok, V., Mpofu, N., & Olusanya, M.** (2021). *Severity estimation of potholes in imagery using convolutional neural networks*. **Journal of Computer Vision Applications, 2**..
4. Abhishek Kumar, Chakrapani Chakrapani, Dhruba Jyoti Kalita and Vibhav Prakash Singh, "A Modern Pothole Detection Technique Using Deep Learning," *Proceedings of the 2nd International Conference on Data, Engineering and Applications*, 2021.
5. Elizabeth Hammell, Alissa McGill, Brian Simmons and Sean Querry, "Pothole Reporting System," Williams Honors College, Honors Research Projects, 2016.